VIDYUTDRISHTI

Technical Architecture & Implementation Guide

AI for Bharat Hackathon · Theme 8 Submission

Theme 8 — AI for Smart Meter Intelligence & Loss Detection by BESCOM

Team Name: Inference Timeout

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1. DATA MODEL & SCHEMA DESIGN

1.1 TimescaleDB Hypertables

The system uses TimescaleDB for time-series data with automatic partitioning. Key design decisions:

```
-- Meter readings hypertable (primary ingestion table)
CREATE TABLE meter_readings (
    time            TIMESTAMPTZ     NOT NULL,
    meter_id        VARCHAR(32)     NOT NULL,
    kwh_consumed    DECIMAL(12,4),
    kvah_consumed   DECIMAL(12,4),
    max_demand_kw   DECIMAL(10,4),
    power_factor    DECIMAL(4,3),
    voltage_l1      DECIMAL(6,2),
    voltage_l2      DECIMAL(6,2),
    voltage_l3      DECIMAL(6,2),
    feeder_id       VARCHAR(16),
    zone_code       VARCHAR(8),
    device_status   INTEGER          -- 0=active  1=comm-fail  2=tamper
);
SELECT create_hypertable('meter_readings', 'time',
    chunk_time_interval => INTERVAL '1 day');
-- Compress chunks older than 7 days
SELECT add_compression_policy('meter_readings', INTERVAL '7 days');
```

1.2 Detection Results Schema

Separate table for detection outputs to enable fast queue queries without scanning time-series data.

```
CREATE TABLE detection_results (
    result_id             SERIAL PRIMARY KEY,
    meter_id              VARCHAR(32)     NOT NULL,
    detected_at           TIMESTAMPTZ     DEFAULT NOW(),
    -- Layer signals
    l0_signal             DECIMAL(4,3),   -- DT energy balance
    l1_zscore             DECIMAL(6,3),   -- Z-score deviation
    l2_deviation          DECIMAL(6,3),   -- Peer comparison
    l3_anomaly_score      DECIMAL(6,4),   -- Isolation Forest
    -- Aggregated confidence
    confidence_score      DECIMAL(4,3),
    confidence_tier       VARCHAR(8),     -- HIGH | MEDIUM | REVIEW | NORMAL
    -- Classification
    pattern_type          VARCHAR(16),    -- sudden_drop | spike | flatline | erratic
    estimated_loss_annual DECIMAL(12,2),
    -- Queue status
    reviewed              BOOLEAN         DEFAULT FALSE,
    reviewer_notes        TEXT,
    feedback_valid        BOOLEAN         -- NULL until reviewed
);
-- Index for prioritised queue queries
CREATE INDEX idx_detection_queue ON detection_results
    (reviewed, confidence_score DESC, estimated_loss_annual DESC)
    WHERE reviewed = FALSE;
```

2. DETECTION ALGORITHM IMPLEMENTATION

2.1 L0: DT Energy Balance (Real-time)

Implemented as a stream processor consuming from Kafka / Redis streams. Runs every 15 minutes with O(1) complexity per feeder.

```
def calculate_l0_signal(feeder_id: str, window_hours: int = 1) -> float:
    """
    L0: Distribution Transformer Energy Balance
    Formula:  |DT injected - sum(meter consumption)| / DT injected
    Threshold: Signal > 0.08 (8%) triggers L0 flag
    """
    dt_injected = get_dt_injection_kwh(feeder_id, window_hours)
    meter_sum = sum(
        reading.kwh
        for meter   in get_feeder_meters(feeder_id)
        for reading in get_readings(meter.id, window_hours)
    )
    imbalance = abs(dt_injected - meter_sum) / dt_injected
    return min(imbalance, 1.0)   # cap at 1.0
```

2.2 L3: Isolation Forest Feature Engineering

12-feature vector extracted per meter. NOT the same as the high-level pitch description.

```
FEATURE_VECTOR = [
    # Consumption pattern   (4 features)
    'hourly_mean', 'hourly_std', 'daily_peak_ratio', 'night_day_ratio',
    # Comparative features  (3 features)
    'peer_zscore', 'feeder_percentile', 'zone_deviation',
    # Temporal patterns     (3 features)
    'dow_pattern_entropy', 'hourly_variance_cv', 'trend_slope_7d',
    # Technical indicators  (2 features)
    'pf_variance', 'voltage_imbalance_max',
]
# Isolation Forest -- parameters tuned via grid search
CONTAMINATION  = 0.03   # expected theft rate
N_ESTIMATORS   = 150
MAX_SAMPLES    = 512
MAX_FEATURES   = 0.8
```

2.3 Confidence Score Aggregation

Weighted combination with sigmoid normalisation for interpretability.

```
import numpy as np

def compute_confidence(l0: float, l1: float, l2: float, l3: float) -> float:
    """
    Weights:  L0=0.10  L1=0.30  L2=0.30  L3=0.30
    Raw score = weighted sum, then sigmoid for 0-1 range.
    """
    weights   = [0.10, 0.30, 0.30, 0.30]
    raw_score = l0*weights[0] + l1*weights[1] + l2*weights[2] + l3*weights[3]
    return round(1 / (1 + np.exp(-5 * (raw_score - 0.5))), 3)

TIERS = {
    'HIGH':   (0.85, 1.00),   # Immediate dispatch
    'MEDIUM': (0.70, 0.85),   # Weekly batch
    'REVIEW': (0.55, 0.70),   # Monthly + pattern check
    'NORMAL': (0.00, 0.55),   # No action
}
```

3. TESTING & VALIDATION FRAMEWORK

3.1 Synthetic Data Injection Strategy

Controlled theft scenario injection for evaluation -- implementation detail not in pitch deck.

Theft Pattern	Injection Method	Detection Target
Meter Bypass (50%)	Drop 40-60% of readings for 5-30 days	L2 peer deviation + L3 pattern
Hook Connection	Reduce consumption by 30% during peak hours only	L1 z-score (time-segmented)
Meter Tamper	Inject PF > 0.99 + voltage imbalance flags	L0 technical validation
Meter Stop	Zero consumption for 48h+ while neighbours consume	L3 flatline detection
Phase Imbalance	Set one phase to 0, others normal	L0 + L3 combined

3.2 Evaluation Harness

Automated comparison against ground-truth labels.

```
class EvaluationHarness:
    def evaluate(self, predictions: List[Detection],
                 ground_truth: List[TheftEvent]) -> Metrics:

        # Match detections to ground truth within 10-day window
        matches = self._temporal_match(predictions, ground_truth, days=10)

        # Per-tier metrics
        for tier in ['HIGH', 'MEDIUM', 'REVIEW']:
            tier preds   = [p for p in predictions if p.tier == tier]
            tier matches = [m for m in matches    if m.prediction.tier == tier]
            precision = len(tier matches) / len(tier preds)
            recall    = len({m.gt for m in tier_matches}) / len(ground_truth)

        # FPR by confidence threshold sweep
        thresholds = np.linspace(0.5, 0.95, 20)
        fpr_by_threshold = {
            t: sum(1 for p in predictions
                   if p.confidence > t and not p.is true positive) / len(predictions)
            for t in thresholds
        }
```

4. SECURITY ARCHITECTURE & OPERATIONS

4.1 Read-Only Integration Model

Critical for BESCOM acceptance -- no control path, only observation.

```
[BESCOM SCADA] ---> [Read-only API Gateway] ---> [VidyutDrishti Backend]
      |                                                  |
      |               [TimescaleDB Replica]                |
      +----------------------------------------------------+
                        (analytics only)

Security Controls
  * API keys with 24h rotation via Vault
  * IP whitelisting (BESCOM data-centre ranges only)
  * No PII in logs (meter id SHA-256 hashed)
  * All queries parameterised (SQL-injection prevention)
  * Read-only DB user -- no INSERT / UPDATE / DELETE on source
```

4.2 Monitoring & Alerting

Metric	Alert Threshold	Response
API latency p95	> 500 ms for 5 min	Scale backend containers
Detection queue backlog	> 100 unreviewed HIGH	PagerDuty to ops team
DB connection pool	> 80% utilisation	Connection limit increase
Isolation Forest inference	> 2s per batch	Model caching enabled
Daily ingestion volume	< 95% of expected	Check AMR data feed

4.3 Backup & Recovery

```
# Automated daily backup (pg_dump) -- runs at 02:00
0 2 * * *  pg_dump -Fc vidyutdrishti > /backup/vd_$(date +%Y%m%d).dump

# Point-in-time recovery (WAL archiving)
# RPO: 15 minutes  |  RTO: 30 minutes (automated failover to replica)

# Retention schedule
#  90 days  -> full detail with meter IDs
#   2 years -> aggregated by zone / feeder (anonymised)
#   5 years -> annual recovery statistics only
```

5. DEPLOYMENT CONFIGURATION

5.1 Docker Compose Production Setup

```
version: '3.8'
services:

  timescaledb:
    image: timescale/timescaledb:latest-pg15
    environment:
      POSTGRES_PASSWORD: ${DB_PASSWORD}
      TIMESCALEDB_TELEMETRY: 'off'
    volumes:
      - tsdb data:/var/lib/postgresql/data
      - ./init-scripts:/docker-entrypoint-initdb.d
    deploy:
      resources:
        limits: { memory: 4G, cpus: '2.0' }

  backend:
    build: ./backend
    environment:
      DATABASE URL:        postgresql://app:${DB PASSWORD}@timescaledb:5432/vidyut
      L3 MODEL PATH:       /models/isolation forest v1.pkl
      L3_BATCH_SIZE:       1000
      CONFIDENCE_CACHE_TTL: 300
    depends on: [timescaledb]
    deploy:
      replicas: 2
      resources:
        limits: { memory: 2G, cpus: '1.0' }

  frontend:
    build:    ./frontend
    ports:    ["5173:80"]
    depends on: [backend]
```

5.2 Environment Variables Reference

Variable	Default	Description
L0_IMBALANCE_THRESHOLD	0.08	DT energy balance alert %
L1_ZSCORE_WINDOW	7d	Rolling window for z-score
L2_PEER_SAMPLE_SIZE	20	Minimum peers for comparison
L3_CONTAMINATION	0.03	Expected theft prevalence
CONFIDENCE_HIGH_MIN	0.85	HIGH tier minimum threshold
FORECAST_HORIZON_DAYS	30	Prophet prediction window
QUEUE_HIGH_MAX_AGE	3	Days before HIGH tier escalates

5.3 Common Issues & Resolution

Symptom	Cause	Fix
L2 peer comparison returns NULL	Insufficient peer meters in zone	Reduce L2_PEER_SAMPLE_SIZE or expand zone
Isolation Forest all 0.5 scores	Model not trained on local patterns	Run POST /admin/train-l3 with 90d history
Frontend shows stale data	Browser caching API responses	Ctrl+Shift+R or check Cache-Control headers
High confidence, no theft found	Technical faults (PF issues) flagged	Check device_status field, add technical filter

6. PERFORMANCE BENCHMARKS

6.1 Load Testing Results (BESCOM-scale simulation)

Tested with 8.5M meter simulation on 4-core / 16GB infrastructure.

Operation	Throughput	Latency (p95)
Meter ingestion	50K readings / sec	45 ms
L0 detection (real-time)	3K feeders / min	12 ms
L1 z-score batch	100K meters / 5 min	180 ms
L2 peer comparison	50K meters / 15 min	450 ms
L3 Isolation Forest	10K meters / batch	2.1 s
Confidence aggregation	25K results / min	8 ms
Queue query (HIGH tier)	500 queries / min	25 ms

6.2 Resource Requirements by Phase

Phase	Meters	Infrastructure Spec
Pilot (3 months)	50,000	4 cores · 8 GB · 50 GB/mo
Zone rollout (12 mo)	500,000	8 cores · 16 GB · 400 GB/mo
Full BESCOM	8,500,000	32 cores · 64 GB · 6 TB/mo

6.3 Query Optimisation Notes

```
-- Continuous aggregate for fast dashboard queries
CREATE MATERIALIZED VIEW daily consumption summary
WITH (timescaledb.continuous) AS
SELECT
    time bucket('1 day', time)  AS bucket,
    meter id,
    avg(kwh consumed)           AS avg daily,
    max(kwh_consumed)           AS peak_daily
FROM meter_readings
GROUP BY bucket, meter_id;

-- Automatic refresh every hour
SELECT add_continuous_aggregate_policy('daily_consumption_summary',
    start offset      => INTERVAL '1 month',
    end offset        => INTERVAL '1 hour',
    schedule_interval => INTERVAL '1 hour');

-- Result: Dashboard loads in < 200ms vs 8s without materialisation
```

7. CONTACT & REPOSITORY

This technical guide accompanies the VidyutDrishti prototype submission for the AI for Bharat Hackathon, Theme 8.

For implementation questions or deployment support:

- Review the full source code in the submitted ZIP file
- API documentation is available at the /docs endpoint when running the backend
- Postman collection included in /tests folder for manual endpoint testing